

CycleGAN for Unpaired Face Sketch-to-Photo Translation

Project Report

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Abstract

This project explores the application of Cycle-Consistent Adversarial Networks (CycleGAN) to the problem of image-to-image translation without paired training data. Specifically, we address the task of converting rough face sketches into realistic facial photographs. Using the CelebAMask-HQ dataset, we train a dual-generator, dual-discriminator architecture implemented in TensorFlow. The model leverages adversarial loss to ensure realism and cycle-consistency loss to ensure the structural integrity of the translation. Despite the constraint of using low-resolution (64×64) inputs due to computational limitations, the model successfully learns the mapping between the sketch and photo domains. The performance is evaluated quantitatively using the Fréchet Inception Distance (FID) score and qualitatively through visual inspection of generated samples. The results demonstrate that CycleGAN is a robust framework for artistic style transfer and domain adaptation even in the absence of ground-truth pairs.

1 1. Introduction

1.1 Problem Definition

Image-to-image translation involves mapping an image from a source domain (e.g., sketches) to a target domain (e.g., photos) while preserving the semantic content of the input. A significant challenge in this field is the scarcity of paired training data—datasets where a specific sketch corresponds exactly to a specific photo are difficult and expensive to curate.

1.2 Motivation and Goal

The ability to generate realistic photos from sketches has profound applications in law enforcement (forensic sketches), entertainment (character design), and digital art tools. The primary goal of this Machine Learning project is to implement a CycleGAN architecture that can learn this translation mapping using *unpaired* datasets. This eliminates the need for one-to-one correspondence between the source and target images during training.

2 2. Literature Review and Related Work

2.1 Generative Adversarial Networks (GANs)

GANs typically consist of a generator and a discriminator competing in a minimax game. While standard GANs can generate realistic images from noise, controlling the output requires conditional inputs.

2.2 Pix2Pix vs. CycleGAN

The Pix2Pix framework serves as a baseline for image translation but strictly requires paired data (e.g., edge map $\leftrightarrow$ photo). In contrast, Zhu et al. (2017) introduced CycleGAN, which introduces a cycle-consistency constraint. This allows the model to learn mappings $G : X \rightarrow Y$ and $F : Y \rightarrow X$ such that $F(G(x)) \approx x$. Our approach utilizes this cycle-consistency property to leverage the unpaired CelebAMask-HQ dataset, making the data collection process significantly easier than methods requiring aligned pairs.

3 3. Problem Definition and Methods

3.1 Formal Definitions

We define two domains:

- **Domain A (Photos):** Realistic face photographs.
- **Domain B (Sketches):** Edge maps generated from photos using edge-detection filters.

The task is to learn a mapping $G : B \rightarrow A$ (Sketch to Photo).

3.2 Architecture

The project utilizes the CycleGAN architecture, which consists of:

1. **Two Generators (G and F):** We employ a ResNet-based generator architecture. This includes an encoder (downsampling), a transformer (residual blocks to maintain structure), and a decoder (upsampling).
2. **Two Discriminators (D_X and D_Y):** We utilize a PatchGAN classifier, which determines whether 70×70 patches of an image are real or fake, forcing the generator to focus on high-frequency details.

3.3 Loss Functions

The objective function is a weighted sum of three losses:

- **Adversarial Loss:** Ensures generated images look like the target domain.
- **Cycle Consistency Loss:** Ensures $F(G(x)) \approx x$ and $G(F(y)) \approx y$.
- **Identity Loss:** Ensures that if a photo is fed into the sketch-to-photo generator, it remains unchanged (optional but helps preserve color composition).

4 4. Implementation Details

4.1 Dataset and Preprocessing

The dataset is derived from CelebAMask-HQ.

- **Filtering:** Images with eyeglasses, hats, or extreme pose angles ($> 20^\circ$) were excluded to ensure consistent facial alignment.
- **Transformation:** All images were resized to 64×64 pixels to accommodate VRAM constraints.

- **Normalization:** Pixel values were scaled to the range $[-1, 1]$.
- **Augmentation:** Random horizontal flipping was applied to increase data diversity.

4.2 Tools and Libraries

The project is implemented using Python, TensorFlow 2.x, and Keras. Key components include 'tf.data' for efficient pipelines and 'tf.GradientTape' for the custom training loop.

4.3 Code Snippet: CycleGAN Training Step

The core logic resides in the custom training step where gradients for both generators and discriminators are calculated simultaneously.

```

1 def train_step(self, batch_data):
2     real_x, real_y = batch_data
3     with tf.GradientTape(persistent=True) as tape:
4         # Generator Forward Pass
5         fake_y = self.gen_F(real_x, training=True)
6         fake_x = self.gen_G(real_y, training=True)
7
8         # Cycle Consistency
9         cycled_x = self.gen_G(fake_y, training=True)
10        cycled_y = self.gen_F(fake_x, training=True)
11
12        # Identity Mapping
13        same_x = self.gen_G(real_x, training=True)
14        same_y = self.gen_F(real_y, training=True)
15
16        # Calculate Total Losses
17        total_cycle_loss = self.cycle_loss_fn(real_x, cycled_x) + \
18                           self.cycle_loss_fn(real_y, cycled_y)
19
20        # ... (Adversarial and Identity loss calculations) ...

```

Listing 1: CycleGAN Training Step Logic

4.4 Custom Layers

To stabilize training, we implemented a custom `InstanceNormalization` layer, which normalizes contrast per image rather than across the batch (as in Batch Normalization), preserving unique stylistic attributes.

5. Experiments and Results

5.1 Experimental Setup

The model was trained for a limited number of epochs to demonstrate feasibility. The input size was fixed at $64 \times 64 \times 3$. We utilized the Adam optimizer with a learning rate of $2e - 4$ and $\beta_1 = 0.5$.

5.2 Visual Results

The generator successfully learned to infer skin texture and facial features from sparse sketch inputs. Qualitative inspection of the results showed that the model could translate edges into

coherent skin tones and facial structures, effectively handling the domain shift between sketches and photos.

5.3 Quantitative Evaluation (FID Score)

We employed the Fréchet Inception Distance (FID) to measure the similarity between the distribution of real photos and generated photos. Using an InceptionV3 backbone for feature extraction:

- **FID Score:** ≈ 0.61

Note: This score was calculated on a small evaluation batch. While it indicates convergence, a full-scale evaluation on the entire test set is recommended for higher precision.

6 6. Discussion and Analysis

6.1 Strengths

The primary strength of this implementation is the successful decoupling of training data. The ResNet-based generator proved effective at maintaining facial geometry, ensuring that the nose and eyes in the generated photo aligned with the sketch.

6.2 Limitations

- **Resolution:** The 64×64 resolution results in blurry output lacking high-frequency skin texture details.
- **Artifacts:** Without extensive training (100+ epochs), checkerboard artifacts remain visible in some translations.
- **Identity Leakage:** In some cases, specific attributes (like hair color) are hallucinated by the model rather than inferred, due to the lack of color information in the sketches.

6.3 Challenges

Computing the FID score resulted in linear algebra warnings regarding singular matrices, suggesting that the batch size for evaluation (100 samples) was relatively small for high-dimensional covariance calculation.

7 7. Conclusion and Future Work

This project demonstrated the capability of CycleGANs to perform sketch-to-photo translation using unpaired data. We successfully built a pipeline comprising data preprocessing, custom GAN architecture, and evaluation metrics.

Future Work:

1. **Higher Resolution:** Scaling the architecture to support 256×256 or 512×512 images for production-grade quality.
2. **Attention Mechanisms:** Implementing Attention-GANs to focus specifically on facial regions (eyes/mouth) for sharper details.
3. **Extended Training:** Training for more epochs to minimize visual artifacts.

8 8. References

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